

Define $\exp: \mathbb{C} \rightarrow \mathbb{C}; z \mapsto \sum_{n=0}^{\infty} \frac{z^n}{n!}$

$\sin: \mathbb{C} \rightarrow \mathbb{C}; z \mapsto \sum_{n=0}^{\infty} (-1)^n \cdot \frac{z^{2n+1}}{(2n+1)!}$

$\cos: \mathbb{C} \rightarrow \mathbb{C}; z \mapsto \sum_{n=0}^{\infty} (-1)^n \cdot \frac{z^{2n}}{(2n)!}$

$$\cos z = \frac{1}{2}(e^{iz} + e^{-iz}), \quad \sin z = \frac{1}{2}(e^{iz} - e^{-iz})$$